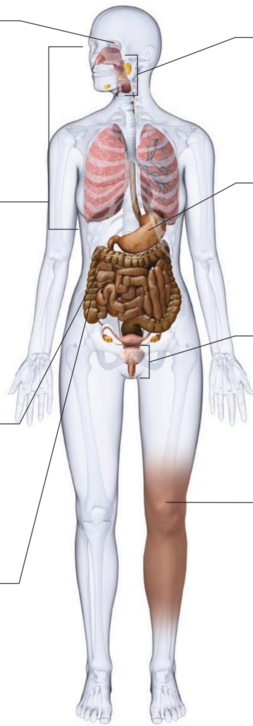
Lumbar lymph nodes
Drain lymph from abdominal organs

AUXILIARY IMMUNE SYSTEM

Many organs have a role in protecting the body against invading microbes. They form an auxiliary immune system that includes the skin, microscopic hairs, gastric enzymes, and useful bacteria.



Tear (lacrimal) glands
Tear fluid contains an antibacterial enzyme, lysozyme, that flushes across the eyeball with each blink

Respiratory tract
Nostril hairs trap airborne particles; mucus and cilia in lining of nose and trachea trap and remove dust, microorganisms, and debris

Small intestine
Digestive enzymes, including those in pancreatic juices, attack microbes that survive the stomach

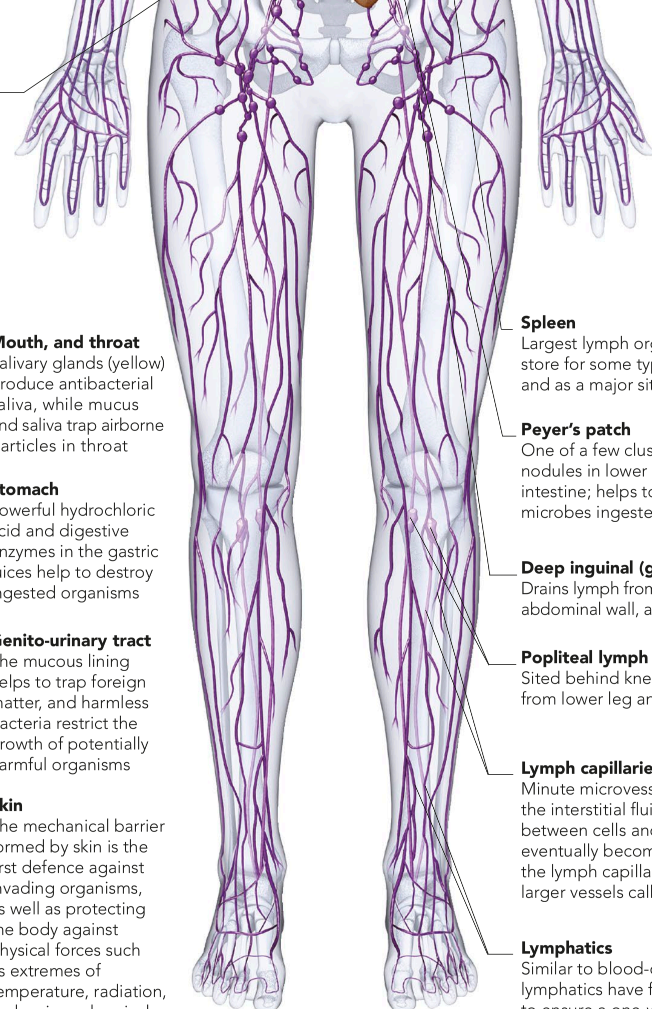
Large intestine
The body's natural gut flora ("friendly" bacteria and other microorganisms) suppress unwanted, harmful microbes

Mouth, and throat
Salivary glands (yellow) produce antibacterial saliva, while mucus and saliva trap airborne particles in throat

Stomach
Powerful hydrochloric acid and digestive enzymes in the gastric juices help to destroy ingested organisms

Genito-urinary tract
The mucous lining helps to trap foreign matter, and harmless bacteria restrict the growth of potentially harmful organisms

Skin
The mechanical barrier formed by skin is the first defence against invading organisms, as well as protecting the body against physical forces such as extremes of temperature, radiation, and various chemicals



Spleen
Largest lymph organ; spleen acts as store for some types of lymphocyte and as a major site for filtering blood

Peyer's patch
One of a few clusters of lymphoid nodules in lower part of small intestine; helps to protect against microbes ingested in food

Deep inguinal (groin) node
Drains lymph from the legs, lower abdominal wall, and external genitals

Popliteal lymph nodes
Sited behind knees; drain lymph from lower leg and foot

Lymph capillaries
Minute microvessels that collect the interstitial fluid, which flows between cells and tissues and eventually becomes lymph fluid; the lymph capillaries unite into larger vessels called lymphatics

Lymphatics
Similar to blood-carrying veins, lymphatics have flap-type valves to ensure a one-way flow of lymph